LIBBY — MANUSCRIPTS INVENTORY

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: every paper reviewed in this case

Generated 2026-08-21

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: 8 clinical (7 included, 1 considered & excluded) + 6 pre-clinical (5 included, 1 considered & excluded) + 38 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

—/— (2026) — ClinicalTrials.gov record (NCT07266298)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** NW-101C (PRAME TCR-engineered T cells)
- **Indication / model:** advanced PRAME-expressing solid tumors, pan-solid basket (NCT07266298) (2L+); Central-lab PRAME-positive AND HLA-A*02:01-positive; ECOG 0-1; RECIST-measurable disease required
- **Design:** first-in-human phase 1 dose-escalation, autologous PRAME TCR-T
- **Effect size:** — — DLT / MTD; ORR
- **Toxicities:** No clinical safety data published. Class expectation for PRAME TCR-T is CRS and ICANS-like neurotoxicity plus lymphodepletion cytopenias, as seen with IMA203 (row osteo-prame-ce-2).
- **Case fit:** partial
- **Notes:** Registry-only; Neowise phase 1 recruiting, no publication or abstract found for NW-101C (searchable name returns no clinical data; the Neowise TCR-T with published data is NW-301V targeting KRAS G12V, a different agent). toxicities[] empty because no source AE data exists. Pan-solid PRAME/HLA-A*02:01 eligibility could admit osteosarcoma if both biomarkers confirm; verify sarcoma is accepted. Efficacy precedent for the PRAME TCR-T class is IMA203 (osteo-prame-ce-2). — (primary endpoint not extracted)

—/— (2026) — ClinicalTrials.gov record (DAREON program)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** obrixtamig
- **Indication / model:** extensive-stage SCLC (1L) (1L); DLL3 (companion DX assay required)
- **Design:** randomized phase 3 (obrixtamig + atezolizumab + carboplatin/etoposide vs SOC)
- **Effect size:** — — OS, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** alveltamig
- **Indication / model:** relapsed SCLC (2L+); DLL3 expression
- **Design:** randomized phase 3 (alveltamig vs topotecan)
- **Effect size:** — — OS, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (FDA Fast Track for ES-SCLC)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** zocilurtatug pelitecan
- **Indication / model:** small-cell lung cancer (2L+); DLL3 expression
- **Design:** randomized phase 3 (zocilurtatug pelitecan vs investigator's choice)
- **Effect size:** — — OS, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SHR-4849 (IDE849)
- **Indication / model:** SCLC, neuroendocrine carcinomas, DLL3-expressing tumors (2L+); DLL3 expression
- **Design:** phase 1/2 ± durvalumab / IDE161
- **Effect size:** — — DLT, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ^{22}Ac-ETN029
- **Indication / model:** SCLC, LCNEC, NEPC, GEP-NEC (2L+); DLL3 expression
- **Design:** phase 1 (^{22}Ac-ETN029 therapy + ^{111}In-ETN029 imaging)
- **Effect size:** — — DLT, MTD, biodistribution
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Hassel/Lorigan (2026) — ClinicalTrials.gov record (PRISM-MEL-301 phase 3)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** brenetafusp (IMC-F106C)
- **Indication / model:** previously untreated HLA-A*02:01-positive advanced cutaneous melanoma (1L); PRAME-positive AND HLA-A*02:01-positive
- **Design:** open-label phase 3 (brenetafusp + nivolumab vs nivolumab ± relatlimab)
- **n:** 670
- **Effect size:** — — PFS, OS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Hong/Wermke (2026) — ClinicalTrials.gov record (SUPRAME phase 3)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA203 (PRAME-TCR-T)
- **Indication / model:** previously treated HLA-A*02:01-positive cutaneous malignant melanoma (2L+); PRAME-positive AND HLA-A*02:01-positive
- **Design:** randomized phase 3 (IMA203 PRAME-TCR-T vs investigator's choice)
- **n:** 360
- **Effect size:** — — PFS, OS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Immatics + Moderna combo)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA203 + mRNA-4203
- **Indication / model:** cutaneous melanoma, synovial sarcoma (2L+); PRAME-positive AND HLA-A*02:01-positive
- **Design:** phase 1 (IMA203 PRAME-TCR-T + mRNA-4203 PRAME mRNA vaccine combination)
- **Effect size:** — — DLT, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Immunocore next-gen ImmTAC)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMC-P115C
- **Indication / model:** PRAME-positive HLA-A*02:01-positive advanced cancer (pan-tumor) (2L+); PRAME-positive AND HLA-A*02:01-positive
- **Design:** phase 1 first-in-human dose-escalation
- **Effect size:** — — DLT, MTD
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [212Pb]Pb-MP0712 (212Pb-DOTAM-MAM279)
- **Indication / model:** SCLC and other DLL3-expressing solid tumors (LCNEC, EP-NEC, GEP-NEC, NEC of bladder) (2L+); DLL3 expression
- **Design:** phase 1/2 theranostic pair: [212Pb]Pb-MP0712 therapy with [203Pb]Pb-MP0712 imaging
- **Effect size:** — — DLT, MTD, dosimetry, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NG2 and DLL3 CAR-T cells
- **Indication / model:** melanoma (2L+); NG2-positive AND DLL3-positive (IHC or flow cytometry)
- **Design:** phase 1/2 autologous dual-target CAR-T
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (SPECTRAL-1)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ML261
- **Indication / model:** R/R SCLC and select NECs (EP-NEC, GEP-NEC, NEPC) (2L+); DLL3 expression
- **Design:** first-in-human phase 1 autologous potency-enhanced DLL3 CAR-T
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DJI136
- **Indication / model:** extensive-stage SCLC (2L+); DLL3 expression (prior DLL3-targeted therapy allowed)
- **Design:** phase 1/2 DLL3-targeted CAR-T
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NEUK203-13 (anti-DLL3 iPSC CAR-NK)
- **Indication / model:** extensive-stage SCLC and neuroendocrine tumors (2L+); DLL3 expression
- **Design:** first-in-human phase 1 iPSC-derived anti-DLL3 CAR-NK with IL-2 support after lymphodepletion
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (NCI rare-cancer study)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** brenetafusp (IMC-F106C)
- **Indication / model:** metastatic/unresectable synovial sarcoma and myxoid/round-cell liposarcoma (2L+);

HLA-A*02:01-positive (PRAME-directed ImmTAC)
- **Design:** phase 2 brenetafusp monotherapy in rare sarcomas
- **Effect size:** — — ORR (RECIST 1.1)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NW-101C (PRAME TCR-T)
- **Indication / model:** advanced solid tumors (pan-solid, PRAME-expressing) (2L+); PRAME-positive (central lab) AND HLA-A*02:01-positive
- **Design:** phase 1 dose-escalation, TCR-engineered T cells after lymphodepletion
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** OPB-201
- **Indication / model:** recurrent endometrial and platinum-resistant ovarian cancer (2L+); PRAME-positive AND HLA-A*02:01 or HLA-A*02:02-positive
- **Design:** phase 1 dose-escalation and expansion
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Mountzios/Rudin (2025) — N Engl J Med

Reference:PMID 40454646 · **Type:** clinical · **Inclusion:** included

- **Intervention:** tarlatamab (DLL3 × CD3 BiTE)
- **Indication / model:** SCLC after platinum-based chemo (2L); Platinum-resistant or platinum-sensitive SCLC; ECOG 0-1
- **Design:** randomized phase 3 (DeLLphi-304)
- **n:** 509
- **Effect size:** 0.6 HR — OS (HR vs chemo)
- **Variance:** 95% CI 0.47–0.77; 95% CI; median OS 13.6 vs 8.3 mo; p=<0.001
- **Toxicities:** CRS (n=254, 56%); CRS G>=3 (n=254, 1%); ICANS-like neurologic event (n=254, 12%); treatment-related AE G>=3 (n=254, 24%); neutropenia G>=3 (n=254, 3%)
- **Case fit:** cross_tumor_only
- **Notes:** Confirmatory phase-3 OS benefit in SCLC. Establishes mechanism convincingly; cross-tumor extrapolation to osteosarcoma still requires DLL3 IHC confirmation.

—/— (2025) — ClinicalTrials.gov record (NCT06814496)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** tarlatamab + radiation therapy (DLL3 × CD3 BiTE + RT)
- **Indication / model:** DLL3-expressing solid tumors, pan-tumor basket with a non-SCLC phase 2 expansion (NCT06814496) (2L+); Histologies with high DLL3 prevalence and an extracranial site amenable to radiation; DLL3 IHC ≥1% gate; nested phase 2 enriches for non-SCLC tumors
- **Design:** first-in-human phase 1/2 single-arm, tarlatamab with concurrent or sequential radiation
- **Effect size:** — — DLT / safety of the RT combination; ORR in the non-SCLC phase 2 expansion
- **Toxicities:** No combination safety data yet. The BiTE component carries the tarlatamab class profile (CRS ~50% mostly G1-2, ICANS-like neurologic events ~10%); the RT arm adds site-specific radiation toxicity that this trial is designed to characterize.
- **Case fit:** partial
- **Notes:** Registry-only; no combination efficacy or safety published (trial recruiting, University of Arizona). Rationale rests on the abscopal hypothesis layered onto tarlatamab monotherapy efficacy (DeLLphi-301/304, rows ce-1/ce-2) and its phase 1 SCLC safety (Paz-Ares/Owonikoko, JCO 2023, PMID 36689692). toxicities[] empty because no combination AE data exists yet: NCT06814496 has no results tab and no abstract. Pan-tumor DLL3-IHC gate could admit an IHC-positive osteosarcoma; confirm bone sarcoma is on the per-protocol list. — (primary endpoint not extracted)

—/— (2025) — ClinicalTrials.gov record / ASCO 2025 abstract TPS2700

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tarlatamab (DLL3 × CD3 BiTE)
- **Indication / model:** DLL3-expressing advanced solid tumors (basket) (2L+); DLL3 IHC ≥25% (stage 1) or ≥1% (stage 2)
- **Design:** single-arm phase 2 basket (Simon two-stage)
- **n:** 29
- **Effect size:** — — ORR at 18 months (RECIST 1.1)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** peluntamig
- **Indication / model:** SCLC, LCNEC, NEPC, GEP-NEC, EP-NEC (2L+); DLL3 expression
- **Design:** phase 1/2 dose-escalation + expansion ± carbo/etoposide / paclitaxel / atezolizumab
- **Effect size:** — — DLT, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record (multi-country IND)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IBI3009

- **Indication / model:** small-cell lung cancer (2L+); DLL3 expression
- **Design:** phase 1 dose-escalation + expansion
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ^{22}Ac-ABD147
- **Indication / model:** SCLC, large-cell neuroendocrine carcinoma (2L+); DLL3 expression
- **Design:** phase 1 dose-finding
- **Effect size:** — — DLT, MTD, biodistribution
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ^{1}Lu-DTPA-SC16.56
- **Indication / model:** neuroendocrine tumors / carcinomas (lung, prostate) (2L+); DLL3 expression
- **Design:** phase 1 (with companion Zr-DFO-SC16.56 PET imaging on NCT04199741)
- **Effect size:** — — DLT, dosimetry, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record (Immatics TCER platform)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA402 (PRAME-TCER)
- **Indication / model:** refractory / recurrent solid tumors expressing PRAME (2L+); PRAME-positive AND HLA-A*02:01-positive
- **Design:** phase 1/2 dose-finding (IMA402 phases Ia / Ib / II)
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tarlatamab + radiation therapy
- **Indication / model:** DLL3-expressing solid tumors (pan-tumor: melanoma, medullary thyroid, sinonasal undifferentiated carcinoma, esthesioneuroblastoma, bladder, testicular, glioblastoma, cervical, LCNEC, NSCLC, Merkel cell) (2L+); DLL3 ≥1% by IHC (histologies with ≥50% prevalence of ≥1% positivity)

- **Design:** phase 1/2 tarlatamab + concurrent or sequential radiation therapy
- **Effect size:** — — DLT, ORR (RECIST 1.1)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BL-M14D1
- **Indication / model:** SCLC and neuroendocrine neoplasms (NEPC, GEP-NEC, Merkel cell, LCNEC, EP-NEC) (2L+); DLL3 expression
- **Design:** phase 1 dose-finding, q3w IV
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record (PRAMETIME-Mel)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PRAME TCR-engineered NK cells
- **Indication / model:** recurrent/refractory melanoma (2L+); PRAME-directed, HLA-A*02:01-positive
- **Design:** phase 1 dose-escalation, PRAME TCR-engineered NK cells with Cy/Flu lymphodepletion
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PRAME TCR-engineered NK cells
- **Indication / model:** relapsed/refractory myeloid malignancies (AML, MDS/CMML) (2L+); PRAME-directed, HLA-A*02:01-positive
- **Design:** phase 1/2 PRAME TCR-engineered NK cells with lymphodepleting chemotherapy
- **Effect size:** — — DLT, MTD, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Hamid/Hassel (2024) — Cancer Discov

Reference:doi:10.1158/2159-8290.CD-24-0298 · **Type:** clinical · **Inclusion:** included

- **Intervention:** brenetafusp (IMC-F106C)
- **Indication / model:** PRAME-positive HLA-A*02:01-positive advanced cutaneous melanoma (2L+); Heavily pretreated melanoma; subset with prior immunotherapy; HLA-A*02:01-positive; PRAME-positive by IHC
- **Design:** open-label phase 1 dose-escalation + expansion (NCT04262466)

- **n:** 132
- **Effect size:** 9 % — ORR; mPFS; mDoR
- **Variance:** 95% CI 4–17; 95% CI in heavily pretreated melanoma; mDoR > 6 mo in responders
- **Toxicities:** CRS (n=132, 85%); CRS G>=3 (n=132, 3%); rash (n=132, 70%); transient hypotension (n=132, 35%); any treatment-related AE G>=3 (n=132, 30%)
- **Case fit:** cross_tumor_only
- **Notes:** First-in-class PRAME × CD3 ImmTAC. Cross-tumor: melanoma data, not osteosarcoma. Establishes the ImmTAC-platform mechanism for PRAME — same fusion-protein architecture as tebentafusp (gp100 × CD3) which has approved indication in uveal melanoma. [Corrected: prior PMID 39007852 was a wrong-identifier bug (resolved to an unrelated breast-cancer paper); nulled per reference-check. Verified brenetafusp Phase-1 data is the ASCO 2024 abstract, DOI 10.1200/JCO.2024.42.16_suppl.9507.]

Wermke/Bauer (2024) — Lancet Oncol

Reference:PMID 40205198 · **Type:** clinical · **Inclusion:** included

- **Intervention:** IMA203 (PRAME-TCR-T)
- **Indication / model:** PRAME-positive HLA-A*02:01-positive refractory solid tumors (basket: melanoma, sarcomas including synovial sarcoma, ovarian, NSCLC, head and neck, uterine, others) (2L+); Heavily pretreated PRAME+ HLA-A*02:01+ patients; subset of synovial sarcoma cohort enrolled; lymphodepleting Cy/Flu prior to infusion
- **Design:** open-label phase 1/2 ACTengine basket (NCT03686124)
- **n:** 28
- **Effect size:** 54 % — ORR (BICR)
- **Variance:** 95% CI 34–73; 95% CI in evaluable cohort; mDoR not reached at 9-mo follow-up
- **Toxicities:** CRS (28/28, 100%); CRS G>=3 (n=28, 11%); ICANS-like neurotoxicity (n=28, 25%); post-Cy/Flu cytopenia G>=3 (28/28, 100%); treatment-related death G5 (1/28, 4%)
- **Case fit:** partial
- **Notes:** Pan-solid PRAME-TCR-T basket including sarcoma cohorts. Best clinical-efficacy signal in the PRAME class to date. Lymphodepletion logistics + CAR-T-style infusion infrastructure required.

Hamid/— (2024) — J Clin Oncol (ASCO 2024 abstract 9507)

Reference:doi:10.1200/JCO.2024.42.16_suppl.9507 · **Type:** clinical · **Inclusion:** included

- **Intervention:** brenetafusp (IMC-F106C, PRAME × CD3 ImmTAC)
- **Indication / model:** post-checkpoint HLA-A*02:01-positive advanced cutaneous melanoma (2L+); HLA-A*02:01-positive unresectable/metastatic cutaneous melanoma; 100% prior anti-PD1, 81% prior anti-CTLA4; efficacy enriched in PRAME-positive subset
- **Design:** phase 1 dose-escalation/expansion monotherapy (IMC-F106C-101, NCT04262466); ASCO 2024 abstract 9507
- **n:** 36
- **Effect size:** 11 % — ORR; DCR; mPFS in PRAME-positive vs PRAME-negative
- **Variance:** 4 PR of 36 RECIST-evaluable at active target doses
- **Toxicities:** CRS; CRS G>=3 (0%); rash
- **Case fit:** cross_tumor_only
- **Notes:** ASCO 2024 abstract — no PubMed PMID; effect sizes from the melanoma cohort, not osteosarcoma. Sarcoma-relevant context is NCI phase 2 NCT07686367 (brenetafusp in synovial sarcoma and

myxoid/round-cell liposarcoma; not-yet-recruiting, no data). toxicities[] carry only what the abstract lists — per-term rates were not tabulated. Complements existing row osteo-prame-ce-1, whose PMID 39007852 is mis-linked (points to a HER2 breast-cancer paper); flagged in the run log.

—/— (2024) — ClinicalTrials.gov record / AACR-NCI-EORTC 2023 abstract

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** gocatamig
- **Indication / model:** DLL3-expressing SCLC, large-cell NEC, neuroendocrine prostate cancer, GEP-NEC (2L+); DLL3 expression
- **Design:** open-label phase 1/2, monotherapy + atezolizumab + ifinatamab deruxtecan combinations
- **Effect size:** — — DLT, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** clesitamig
- **Indication / model:** SCLC, neuroendocrine carcinoma (2L+); DLL3 expression
- **Design:** phase 1 dose-finding ± tocilizumab
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FZ-AD005
- **Indication / model:** advanced solid tumors, SCLC, LCNEC (2L+); DLL3 expression
- **Design:** phase 1 dose-finding
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LB2102
- **Indication / model:** extensive-stage SCLC, LCNEC of lung (2L+); DLL3 expression
- **Design:** phase 1 autologous CAR-T dose-finding
- **Effect size:** — — DLT, MTD, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Hamid/Hassel (2024) — ClinicalTrials.gov record / ASCO 2024 abstract

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** brenetafusp (IMC-F106C)
- **Indication / model:** advanced solid tumors expressing PRAME (cutaneous and uveal melanoma, ovarian, lung, sarcoma cohorts) (2L+); PRAME-positive AND HLA-A*02:01-positive
- **Design:** phase 1/2 brenetafusp monotherapy + combinations (pembrolizumab, chemo, tebentafusp, bevacizumab, kinase inhibitors)
- **Effect size:** — — DLT, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wermke/Bauer (2024) — ClinicalTrials.gov record / Lancet Oncol 2024 (ACTengine phase 1)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA203 (PRAME-TCR-T) / IMA203CD8
- **Indication / model:** refractory / recurrent solid tumors expressing PRAME (basket: melanoma, sarcomas including synovial sarcoma + osteosarcoma, ovarian, NSCLC, head and neck, uterine, others) (2L+); PRAME-positive (IHC) AND HLA-A*02:01-positive
- **Design:** open-label phase 1/2 (IMA203 ± IMA203CD8 product) — ACTengine pan-solid basket
- **Effect size:** — — DLT, ORR, MTD
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Ahn/Paz-Ares (2023) — N Engl J Med

Reference:PMID 37861218 · **Type:** clinical · **Inclusion:** included

- **Intervention:** tarlatamab (DLL3 × CD3 BiTE)
- **Indication / model:** previously treated SCLC (2L+); SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated
- **Design:** open-label phase 2 dose-evaluation (DeLLphi-301)
- **n:** 220
- **Effect size:** 40 % — ORR (BICR)
- **Variance:** 95% CI 29–52; 95% CI in 10mg cohort
- **Toxicities:** CRS (n=100, 51%); CRS G>=3 (n=100, 1%); ICANS-like neurologic event (n=100, 10%); treatment-related AE G>=3 (n=100, 30%); treatment discontinuation due to AE (n=100, 3%)
- **Case fit:** cross_tumor_only
- **Notes:** Pivotal trial driving FDA accelerated approval. Cross-tumor row: patient has osteosarcoma, not SCLC — included as the load-bearing efficacy evidence the basket trial NCT06788938 extrapolates from.

Zhang/Shi (2023) — Aging (Albany NY)

Reference:PMID 37179118 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DLL3 pan-cancer expression atlas
- **Indication / model:** TCGA pan-cancer transcriptomic + protein-atlas analysis
- **Design:** Computational characterization of DLL3 expression across tumor types and correlations with TMB, MSI, and tumor microenvironment.
- **n:** 33 TCGA tumor types, ~10,000 samples
- **Effect size:** moderate — DLL3 expression detectable across 18 cancer types; co-varies with immune infiltration in subsets. Sarcoma signal present but not membrane-protein-validated.
- **Variance:** vs GTEx normal-tissue baseline; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Bulk RNA-level analysis. Surface-protein expression — the requirement for BiTE/ADC engagement — is not interrogated. Useful for hypothesis generation; insufficient to gate clinical decisions.

—/— (2023) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** QLS31904
- **Indication / model:** advanced solid tumors with DLL3 expression (2L+); DLL3 expression
- **Design:** phase 1 dose-finding
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2023) — ClinicalTrials.gov record

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DLL3-CAR-NK cells
- **Indication / model:** extensive-stage SCLC (2L+); DLL3 expression
- **Design:** phase 1 allogeneic DLL3-CAR-NK cell infusion
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Yao/Pavel (2022) — Oncologist

Reference:PMID 35983951 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DLL3 as therapeutic target — review
- **Indication / model:** Literature review
- **Design:** Comprehensive overview of DLL3 biology, IHC characterization, and therapeutic-development landscape (BiTEs, ADCs, CAR-T, radio-conjugates).
- **Effect size:** moderate — DLL3 is a 'tumor-restricted' antigen with low normal-tissue expression — attractive when present at adequate density. Surface-protein detection (e.g. SP347 IHC) is the consensus gating biomarker for DLL3-directed therapies.
- **Variance:** vs —; translatability: med
- **Toxicities:** n/a (preclinical)

- **Case fit:** weak
- **Notes:** Focuses on neuroendocrine neoplasms. Does not establish osteosarcoma-specific evidence. Reinforces the protein-IHC requirement that the user's 'DLL3 RNA' alone does not satisfy.

Blackhall/Carbone (2021) — JTO

Reference:PMID 33508456 · **Type:** clinical · **Inclusion:** excluded — Drug discontinued by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism (DLL3-targeted PBD-warhead ADC) is informative for the target rationale but the molecule itself is not procurable.

- **Intervention:** rovalpituzumab tesirine (Rova-T, DLL3 ADC)
- **Notes:** Excluded: Drug discontinued by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism (DLL3-targeted PBD-warhead ADC) is informative for the target rationale but the molecule itself is not procurable.

Hipp/Wolf (2020) — Clin Cancer Res

Reference:PMID 32694156 · **Type:** preclinical · **Inclusion:** excluded — Original BiTE construct (AMG 757 / tarlatamab) characterization in SCLC models — superseded by Giffin 2021 (pmid:35983951) which includes the DLL3-density-dependent cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.

- **Intervention:** tarlatamab (DLL3 × CD3 BiTE)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Original BiTE construct (AMG 757 / tarlatamab) characterization in SCLC models — superseded by Giffin 2021 (pmid:35983951) which includes the DLL3-density-dependent cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.

—/— (2020) — ClinicalTrials.gov record (Bellicum legacy program)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BPX-701
- **Indication / model:** AML, MDS, uveal melanoma (2L+); PRAME-positive
- **Design:** phase 1/2 BPX-701 PRAME-TCR with iCasp9 safety switch + rimiducid
- **Effect size:** — — DLT, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Giffin/Coxon (2019) — Clin Cancer Res (developmental program)

Reference: — · **Type:** preclinical · **Inclusion:** included

- **Intervention:** tarlatamab (DLL3 × CD3 BiTE)
- **Indication / model:** DLL3-positive SCLC xenograft + humanized PBMC co-culture
- **Design:** Bispecific antibody co-engages DLL3 on tumor cells and CD3 on T cells, redirecting cytotoxic T-cell killing.
- **n:** n=8 mice/arm; in vitro replicates n=3
- **Effect size:** strong — DLL3-density-dependent cytolysis demonstrated; ≥1% surface DLL3 by IHC sufficient for

activity in SCLC models. Tumor regression sustained beyond treatment cessation in xenograft.
- **Variance:** vs vehicle + PBMCs without bispecific; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** All published mechanistic data are SCLC-derived. No primary preclinical osteosarcoma data with tarlatamab in the public literature. Translatability to osteosarcoma depends on whether DLL3 is membrane-localized and density-sufficient in osteosarcoma cells — currently unknown.

Wang/Zhu (2018) — J Exp Clin Cancer Res

Reference:PMID 29475441 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Notch3/DLL3 pathway in osteosarcoma (target rationale)
- **Indication / model:** Osteosarcoma cell lines (MG-63, U2OS, Saos-2); primary osteosarcoma samples
- **Design:** Notch3-DLL3 axis implicated in osteosarcoma stemness and chemoresistance; SOX2OT lncRNA upregulates Notch3/DLL3 signaling.
- **n:** n=3 biological replicates per condition
- **Effect size:** moderate — EGCG + doxorubicin combination suppresses Notch3/DLL3 signaling and reduces stemness in osteosarcoma; suggests pathway is operative, not just expressed.
- **Variance:** vs vehicle + scrambled siRNA / parental cell line; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Pathway-level mechanistic study; does NOT measure cell-surface DLL3 protein expression by IHC. Does not address whether DLL3 is on the membrane (required for BiTE/ADC engagement) vs intracellular only. The user's 'DLL3 RNA expression' input matches the type of evidence here — but this work cannot bridge the RNA-to-membrane-protein gap that DLL3-targeted therapies need.

—/— (2018) — ClinicalTrials.gov record (Amgen pipeline)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** AMG 119
- **Indication / model:** small-cell lung cancer (2L+); DLL3 expression
- **Design:** phase 1 autologous CAR-T (legacy first-generation DLL3 CAR-T)
- **Effect size:** — — DLT, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2018) — ClinicalTrials.gov record (Stemcentrx pipeline; pre-AbbVie acquisition)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SC-002
- **Indication / model:** small-cell lung cancer (2L+); DLL3 expression
- **Design:** phase 1 (legacy second-generation DLL3 ADC)
- **Effect size:** — — DLT, MTD
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Iura/Yoshida (2017) — Hum Pathol

Reference:doi:10.1016/j.humpath.2017.03.010 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRAME expression in osteosarcoma (target rationale)
- **Indication / model:** Human osteosarcoma tumor specimens (n=82); PRAME IHC staining
- **Design:** PRAME is a cancer-testis antigen with restricted normal-tissue expression (testis) and broad solid-tumor expression. In osteosarcoma, PRAME IHC is positive in ~50-60% of cases, with expression intensity varying by histologic subtype.
- **n:** n=82 specimens
- **Effect size:** moderate — PRAME IHC positivity in 56% of osteosarcoma cases (46/82); higher rates in osteoblastic and chondroblastic subtypes; correlation with metastatic phenotype.
- **Variance:** vs non-osteosarcoma sarcoma comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** RNA-protein concordance is imperfect; protein-level confirmation by IHC is the load-bearing step before any PRAME-targeting therapy. HLA-A*02:01 typing additionally required for ImmTAC and TCR-T programs.